

Al-Khwarizmi 2024 Day 2:

Solutions and Notes

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Structure

Each problem is accompanied by

1. **Hints:** Guiding questions (ideally to be read one by one) if you are stuck while attempting the problem.
2. **Thought processes/Motivation:** A walkthrough of how I arrived at a solution.
3. **Solution:** A write-up in a style similar to what I would submit in a contest.

Things to note in a solution write-up:

1. In problems that require you to find an answer (e.g. find all x such that ...), be sure to clearly state it (even if you don't have a complete proof). Ideally, state the answer at the very beginning of the write-up (though I didn't do that here for the sake of avoiding redundancy).
2. In the write-up, you can omit all thought processes and motivation—this is recommended to save time.

1 Problem 1

At a party, every guest is a friend of exactly fourteen other guests (not including him or her). Every two friends have exactly six other attending friends in common, whereas every pair of non-friends has only two friends in common. How many guests are at the party? Please explain your answer with proof.

Hints:

- How can you reinterpret this problem as a drawing?
- Try double counting. What value should you double count?
- To be more specific, what triple should you double count?
- Can you think of a scenario where the statement is realised?

Thought processes/Motivation:

- Whenever we have some relation between two things (in this case, friendship), it is always a good idea to think about how we can reinterpret the problem as either a **graph** or **matrix**. Suppose each guest is represented by a vertex in a simple graph, and two vertices are connected if and only if the corresponding guests are friends. Then,
 - each vertex has degree 14,
 - every pair of adjacent vertices has 6 common neighbours, and
 - every pair of non-adjacent vertices has 2 common neighbours.
- This problem screams **double counting** (refer to another resource if unfamiliar)—but what value should we double count? We realise that counting pairs does not make sense, so we should attempt to count triples.
- What triple should we count? In reality, it should not matter too much as long as we know how to work it out, but let's attempt to double count the number of triples so that one guest is friends with the other two. In other words, we wish to count triples of vertices (x, y, z) so that x is adjacent to both y and z .

- Suppose there are k guests. Fixing x , there are $\binom{14}{2}$ choices of the pair (y, z) . Since there are k possible vertices x , the desired value is $k\binom{14}{2}$.
- On the other hand, what if we fix the pair (y, z) ? If y and z are adjacent, there are 6 choices of x . Since there are $7k$ such pairs of (y, z) , which follows from the fact there are a total of $\frac{14k}{2}$ edges, there are $7k \times 6$ triples where y and z are adjacent. If y and z are not adjacent, there are 2 choices of x . Since there are $\binom{k}{2} - 7k$ such pairs of (y, z) , there are $(\binom{k}{2} - 7k) \times 2$ triples where y and z are not adjacent.
- Therefore, we obtain the equality

$$k\binom{14}{2} = (7k \times 6) + ((\binom{k}{2} - 7k) \times 2).$$

Solving this gives us $k = 64$.

- Indeed, this is achievable if each guest is represented as a cell in an 8×8 grid, in which guests are considered friends if and only if they belong to the same row or column.

Solution:

Represent the conditions as a graph and let there be k guests. We will count the number of triples of vertices (x, y, z) so that x is adjacent to both y and z . Fixing x , there are $k\binom{14}{2}$ triples, while fixing (y, z) , there are $(7k \times 6) + ((\binom{k}{2} - 7k) \times 2)$ triples. Therefore,

$$k\binom{14}{2} = (7k \times 6) + ((\binom{k}{2} - 7k) \times 2) \implies k = 64.$$

Indeed, this is achievable if each guest is represented as a cell in an 8×8 grid, in which guests are considered friends if and only if they belong to the same row or column. ■

2 Problem 2

Let a, b, c be distinct real numbers such that $a + b + c = 0$ and

$$a^2 - b = b^2 - c = c^2 - a.$$

Evaluate all the possible values of $ab + ac + bc$.

Hints:

- Are you reminded of any algebraic identities? How can you make use of them here?
- To be more specific, do the squares remind you of any identity? What about $a + b + c = 0$?
- Can you obtain the values of $(a + b)(b + c)(c + a)$, abc or $a^3 + b^3 + c^3$?

Thought processes/Motivation:

- Note that there are usually multiple solutions to this kind of problem, but here, I will present the thought process to mine. As a rule of thumb, problems like this (i.e. problems which ask you to evaluate the value of some expression) require **algebraic manipulations** to be solved, meaning that we are expected to manipulate the given equations in order to arrive at the value of the desired expression. This means we are NOT expected (nor recommended) to solve for each variable, then substitute those values to calculate the desired expression (though yes, in some cases this may be possible, but it will be extremely ugly).
- So generally, a good starting point for this type of problem is to first think about **algebraic identities** which could be useful. For example, one of the first algebraic identities that I thought about was

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac)$$

because we have the value of $a + b + c$ and we want to find $ab + bc + ca$.

- Observing the given equality, this immediately reminds us of the difference of squares identity. Furthermore, with enough experience, we

know that this results in something nice because we have repeated expressions that can be cancelled out. In particular, we can rewrite the condition as

$$(a+b)(a-b) = b-c$$

$$(b+c)(b-c) = c-a$$

$$(c+a)(c-a) = a-b$$

Since we know a, b, c are pairwise distinct, we know this means

$$a-b, b-c, c-a \neq 0$$

so multiplying the 3 equations gives

$$(a+b)(b+c)(c+a)(a-b)(b-c)(c-a) = (a-b)(b-c)(c-a)$$

$$\implies (a+b)(b+c)(c+a) = 1$$

$$\implies -(-c)(-a)(-b) = abc = -1$$

Take note that we divided $(a-b)(b-c)(c-a)$ since we are sure it is non-zero. In general, always be careful when dividing.

- Moreover, it is a well-known lemma that

$$a+b+c=0 \implies a^3+b^3+c^3=3abc$$

which follows from the identity

$$a^3+b^3+c^3 = (a+b+c)(a^2+b^2+c^2-ab-bc-ac) + 3abc.$$

So, we now know that

$$a^3+b^3+c^3 = -3.$$

- This final step is arguably the trickiest step. Notice that

$$a^3+b^3+c^3-ab-bc-ca = a(a^2-b)+b(b^2-c)+c(c^2-a) = aX+bX+cX = 0.$$

Therefore,

$$ab+bc+ca = a^3+b^3+c^3 = -3,$$

as desired.

Short note: You may think many steps (for example, the previous step) seem unmotivated and unintuitive. In reality, that is true for many problems involving algebraic manipulations like this one. However, intuition is developed through lots of practice and experience. For instance, I knew that $a+b+c=0$ might hint at $a^3+b^3+c^3=3abc$ only through experience. Without experience, I agree that this step is quite unintuitive.

Solution:

The given condition implies

$$(a+b)(a-b) = b-c$$

$$(b+c)(b-c) = c-a$$

$$(c+a)(c-a) = a-b$$

Multiplying these equations (and keeping in mind that $(a-b)(b-c)(c-a) \neq 0$), we obtain

$$(a+b)(b+c)(c+a)(a-b)(b-c)(c-a) = (a-b)(b-c)(c-a)$$

$$\implies (a+b)(b+c)(c+a) = 1$$

$$\implies -(-c)(-a)(-b) = abc = -1.$$

Finally,

$$a^3 + b^3 + c^3 - ab - bc - ca = a(a^2 - b) + b(b^2 - c) + c(c^2 - a) = X(a+b+c) = 0$$

$$\therefore ab + bc + ca = a^3 + b^3 + c^3 = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac) + 3abc = -3. \quad \blacksquare$$

3 Problem 3

Two circles with centers O_1 and O_2 intersect at P and Q . Let ω be the circumcircle of the triangle PO_1O_2 ; the circle ω intersect the circles centered at O_1 and O_2 at points A and B , respectively. The point Q is inside triangle PAB and PQ intersects ω at M . The point E on ω is such that $PQ = QE$. Let ME and AB meet at L , prove that

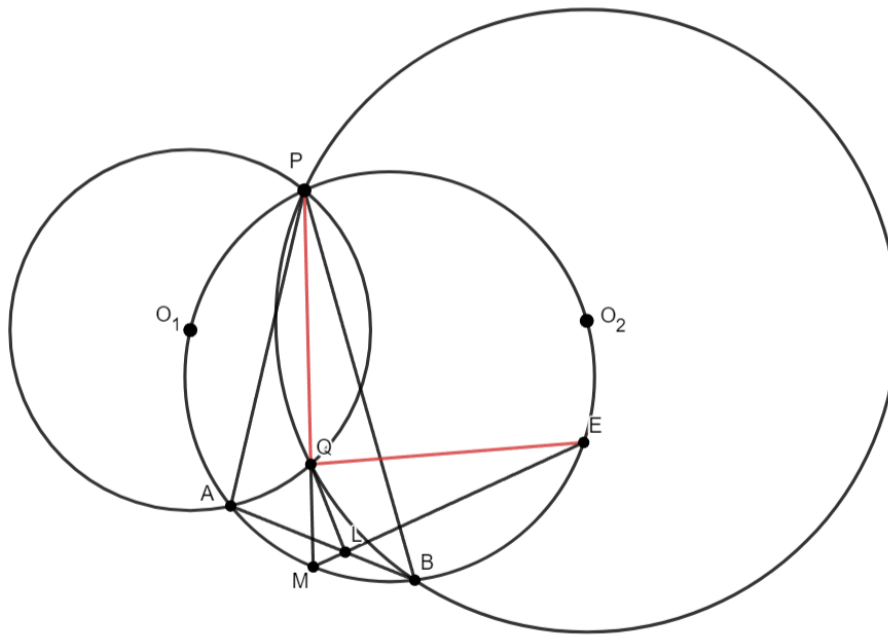
$$\angle QLA = \angle MLA.$$

Hints:

- Do any triples of points seem collinear? Are there any lengths that seem equal?
- SPOILER: Can you prove Q lies on AO_2 and BO_1 ? How does this imply that M is the midpoint of arc AB ?
- Work backwards. What is the problem equivalent to?
- Are you able to angle chase $\angle QLA$ or $\angle MLA$? How? And what does this imply, assuming the problem statement is true?

Note about Olympiad geometry problems:

- One general approach to solve geometry problems is as follows:
 Stare at drawn diagram $\rightarrow$ Make conjecture based on diagram
 $\rightarrow$ Prove or disprove conjecture $\rightarrow$ Repeat
- Conjectures are **any guesses** you might make about a certain geometric configuration. Examples include conjecturing
 - points A , B and C are collinear
 - quadrilateral $ABCD$ is a cyclic quadrilateral
 - line AB is parallel to line CD
 - etc.—Conjectures can literally be anything



Thought processes/Motivation:

- Note: Familiarity with angle chasing (basics + for circles) and the Power of a Point Theorem is assumed.
- Observe the diagram drawn. We begin to come up with some **conjectures**. For example, one of the first things we might conjecture is that point Q lies on AO_2 and BO_1 . It also seems like $MA = MB$. In fact, maybe $MA = MB = MQ$?
- Let's try to prove our first conjecture: Q lies on AO_2 ? Indeed, this is true since

$$\angle O_2AP = \angle O_2O_1P = \frac{\angle QO_1P}{2} = \angle QAP.$$

Symmetrically, this must also mean that Q lies on BO_1 .

- Then, we realise that this means Q is the incenter of $\triangle PAB$ (because AQ and BQ are angle bisectors since $O_1A = O_1P$ and $O_2B = O_2P$), which actually implies our second conjecture, i.e. $MA = MB$. Furthermore, the well-known Incenter-Excenter Lemma (search this up yourself) confirms that $MA = MB = MQ$. (This can also be easily proven via an easy angle chase, but it is always useful to know lemmas like this one.)
- Looking at our diagram, we fail to see any more obvious potential properties (e.g. collinear points). So, we can start to **work backwards**. Recall that we wish to show that $\angle QLA = \angle MLA$. Considering these two angles, the angle that seems easier to angle chase is $\angle MLA$, so let's try doing that. Also, let's set $\angle MPA = \angle MPB = x$ and $\angle QPE = \angle QEP = y$ to make our work simpler. Then, we obtain that

$$\begin{aligned}
 \angle MLA &= 180^\circ - \angle AML - \angle ALM \\
 &= 180^\circ - (180^\circ - \angle APE) - x \\
 &= 180^\circ - (180^\circ - x - y) - x \\
 &= y.
 \end{aligned}$$

- Hence, by the problem statement, we know to expect that $\angle QLA = y$. This means

$$\angle QLE = 180^\circ - 2y = \angle PQE.$$

Notice that this is true if and only if MQ is tangent to the circumcircle of $\triangle QLE$! So instead, we aim to prove this **equivalent statement**.

- We try to angle chase this directly, but it does not seem fruitful. Recall we also know that $MA = MB = MQ$. Hence, this motivates us to apply the Power of a Point theorem, so we wish to prove that

$$MQ^2 = ML \times ME.$$

Since $MA = MQ$, it is enough to show that

$$MA^2 = ML \times ME,$$

which is true if and only if MA is tangent to the circumcircle of $\triangle ALE$. And this is true by a trivial angle chase because clearly,

$$\angle MEA = \angle MAL$$

since $MA = MB$. Therefore, we are done!

- Essentially, what we did was first scout the diagram and proved some conjectures. Once we were stuck, we started working backwards to discover some less obvious properties, which can be proven with the facts we already know.
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Solution:

Q lies on AO_2 because

$$\angle O_2AP = \angle O_2O_1P = \frac{\angle QO_1P}{2} = \angle QAP.$$

Furthermore, $O_2B = O_2P$, so AQ bisects $\angle PAB$. Symmetrically, Q lies on BO_1 , and BQ bisects $\angle PBA$. Thus, Q is the incenter of $\triangle PAB$ and $MA = MB = MQ$ by the Incenter-Excenter Lemma. Next, notice that

$$\begin{aligned} \angle MLA &= 180^\circ - \angle AML - \angle ALM \\ &= 180^\circ - (180^\circ - \angle APE) - \angle MPB \\ &= 180^\circ - (180^\circ - \angle MPA - \angle MPE) - \angle MPA \\ &= \angle MPE. \end{aligned}$$

Finally, observe that MA is tangent to the circumcircle of $\triangle ALE$ because

$$\angle MEA = \angle MAL$$

since $MA = MB$. Thus,

$$MQ^2 = MA^2 = ML \times ME,$$

implying that MQ is tangent to the circumcircle of $\triangle QLE$. Therefore,

$$\angle QLM = \angle MQE = 2\angle MPE = 2\angle MLA. \quad \blacksquare$$